

Beyond the Mean: A Weighted k -Sample Omnibus Variance-Ratio Statistic for Covariate Balance Diagnostics

Ariel Linden, DrPH

University of California, San Francisco

Department of Medicine

Division of Clinical Informatics & Digital Transformation (DoC-IT)

San Francisco, CA, USA

ariel.linden@ucsf.edu

Abstract

Rubin’s variance ratio (VR) complements the standardized mean difference by detecting covariate imbalance in spread, but no purpose-built omnibus extension exists for more than two groups. We introduce F_{VR} , a size-weighted quadratic combination of pairwise log-variance-ratios that generalizes VR to k groups, together with geometric-mean and maximum pairwise variance-ratio comparators. F_{VR} has an exact relationship to Rubin’s VR at $k = 2$ and a mathematical structure directly parallel to Cohen’s f . In Monte Carlo simulations spanning four variance-driven bias mechanisms, $k = 3, 4, 6$, and sample sizes from 200 to 10,000, F_{VR} and the geometric-mean statistic generally tracked downstream estimation bias at least as well as the maximum statistic. F_{VR} retained substantially more signal when many

groups shared a very small sample, while absolute-bias thresholds were unstable at $N = 200$. Across mechanisms, $F_{VR} < 0.10$ was reasonably reassuring, values above approximately 0.30 generally indicated concern, and intermediate values were context-dependent. The statistic is implemented for arbitrary analysis weights in the Stata command `varatio` and illustrated using a multivalued-treatment application.

Keywords

variance ratio; covariate balance; effect size; multivalued treatments; simulation study

1 Introduction

Covariate balance is a necessary step in supporting a causal interpretation of an estimated treatment effect, achieved by design, as in randomized experiments, or by adjustment, as in propensity-score weighting (Robins et al., 2000), marginal mean weights through stratification (Linden, 2014), entropy balancing (Hainmueller, 2012), or matching (Linden and Samuels, 2013). Demonstrating balance on observed characteristics shows that an adjustment procedure has balanced the variables it was given; it does not, by itself, guarantee balance on unmeasured confounders, on which any causal interpretation also depends (Rubin, 2008).

For a continuous covariate compared between two groups, balance is conventionally assessed on (at least) two moments simultaneously: the standardized mean difference (SMD), and the variance ratio (VR) introduced by Rubin (Rubin, 2001), who showed that groups can share an acceptable SMD while differing enough in variance that linear regression adjustment grossly over- or under-corrects for the remaining bias. That finding traces to an explicitly empirical source: Rubin’s own $VR > 2$ (or < 0.5) convention is drawn directly from simulation evidence in Cochran and Rubin (Cochran and Rubin, 1973), whose Table 1 — reproduced in Rubin’s 2001 paper — shows regression-adjustment bias reduction collaps-

ing or reversing sign once the variance ratio between two groups reaches 2 or its reciprocal. The threshold marks an empirically located point of adjustment failure, not an arbitrary round number.

When more than two groups are compared, two very different extensions of this two-group picture now exist. For the mean, Cohen’s f generalizes the SMD to $k > 2$ groups with an exact, derivable relationship to the two-group case (Linden, 2026a). For the entire distribution, the weighted k -sample Kolmogorov–Smirnov, Cramér–von Mises, and Anderson–Darling tests extend nonparametric distributional comparison to $k > 2$ groups, again reducing exactly to their two-group counterparts at $k = 2$ (Linden, 2026c). Variance imbalance, however, sits in a gap between these two extremes: a specific higher moment, more than the mean captures but far short of the full distribution these nonparametric tests characterize, and no purpose-built k -sample omnibus statistic for it exists. What is available instead is ad hoc: software such as the R package `cobalt` extends Rubin’s pairwise VR to multi-category treatments by reporting the single largest pairwise ratio, following the “Max2SB” maximum-aggregation logic developed by Lopez and Gutman (Lopez and Gutman, 2017) for standardized mean bias specifically, not for variance ratios; the extension to variance is an unpublished software design choice, not an independently validated statistic in its own right.

This paper takes up that gap. We ask the same comparative question already posed and answered for the mean (Linden, 2026a): which k -sample statistic best summarizes variance-ratio imbalance — the existing maximum-based aggregation, its natural mean-based counterpart, or a Cohen’s- f -style quadratic combination? We then ask whether the answer depends on how imbalance actually translates into downstream estimation bias, rather than on an abstract target alone. We introduce F_{VR} , a size-weighted quadratic combination of pairwise log-variance-ratios, alongside two comparators generalized to the same setting: GMVR, the geometric mean of the pairwise ratios, and the maximum pairwise ratio itself, the statistic `cobalt` already reports. All three are implemented for arbitrary weighted models via the

community-contributed Stata command `varatio` (Linden, 2026b). Although much of the motivation above centers on propensity-score adjustment, the underlying construction (Section 2.1) accepts any observation-level analysis weights and is not specific to generalized-propensity-score or inverse-probability of treatment weighting (IPTW): it applies equally to balance assessment after entropy balancing, calibration weighting, matching weights, or marginal mean weighting through stratification (MMWS), or any other procedure that produces analysis weights.

The paper’s contributions fall into three categories. The mathematical contribution is twofold. First, we derive F_{VR} ’s exact relationship to Rubin’s VR at $k = 2$ (Section 2.3). Second, we show that the minimum-ratio result proven for Cohen’s f (Linden, 2026a) transfers, without modification, to the log-variance scale (Section 2.5): the same underlying algebra applies to any set of k real numbers, whether they represent group means or log-variances, so the proof did not need to be redone, only relabeled. The computational contribution is `varatio` itself, implementing F_{VR} , GMVR, and the maximum pairwise ratio for arbitrary weighted models, with a weighted-variance point estimate that remains correct regardless of whether weights correlate with the covariate being assessed. The empirical contribution is a Monte Carlo study spanning four structurally distinct bias-generating mechanisms, sample sizes from 200 to 10,000, and $k \in \{3, 4, 6\}$, together with a classification-accuracy analysis that identifies a preliminary, evidence-based interpretive range for F_{VR} , following the same simulation-derived approach Cochran and Rubin (Cochran and Rubin, 1973) originally used to establish the two-group convention this paper extends.

2 Methods

2.1 Notation and General Framework

Consider a continuous covariate X and a treatment variable with $k \geq 2$ levels, denoted Groups $1, \dots, k$, with n_j observations in Group j and $N = \sum_j n_j$. Unlike the mean, which is

naturally obtained as a regression coefficient or adjusted predictive margin, variance has no equally direct model-based analog available across arbitrary model types; accordingly, each group’s variance is computed directly as the weighted second central moment, normalized by total analysis weight,

$$s_j^2 = \frac{\sum_i w_{ji}(x_{ji} - \bar{x}_j^w)^2}{\sum_i w_{ji}}, \quad \bar{x}_j^w = \frac{\sum_i w_{ji}x_{ji}}{\sum_i w_{ji}}, \quad (1)$$

using whatever observation-level weights w_{ji} the analysis specifies — for instance, inverse-probability or MMWS weights — and reducing to the ordinary empirical second central moment (variance with divisor n , not $n - 1$) when all weights equal 1. A direct consequence worth noting: if $X^* = aX + b$ for any $a \neq 0$, then $s_j^{*2} = a^2 s_j^2$, so $\ell_j^* = \log(a^2) + \ell_j$ — the same additive constant for every group, which cancels exactly in every deviation $\ell_j - \bar{\ell}$ used below. F_{VR} , and every pairwise ratio built from it, is therefore exactly invariant to any location shift or nonzero linear rescaling of X ; converting a covariate from dollars to thousands of dollars, or kilograms to pounds, cannot change the balance diagnosis. We avoid the term “standard weighted sample variance” since several competing definitions exist, particularly regarding finite-sample corrections and effective degrees of freedom; the quantity above is used as a balance diagnostic, not as a claim to any particular unbiased-estimator property under a specific sampling model. This differs from the model-margin approach used for the mean-based statistics of Linden (2026a). The direct formula above remains valid when the observation-level weights correlate with the covariate being assessed, as can occur under inverse-probability weighting. Section 2.3 introduces a second, distinct set of weights, $w_j = n_j/N$, used only in aggregating across groups; the two serve different roles and are not interchangeable, a distinction made explicit there.

2.2 The Two-Group Case: Rubin’s Variance Ratio

For $k = 2$ groups, Rubin’s (Rubin, 2001) variance ratio is

$$VR = \frac{s_1^2}{s_0^2}, \tag{2}$$

with values near 1 indicating balance and values above 2 or below 0.5 indicating imbalance severe enough that linear regression adjustment can grossly over- or under-correct for the remaining bias (Cochran and Rubin, 1973). Because VR depends on which group’s variance is larger, we follow Rubin’s own symmetrized construction,

$$VR^* = \frac{\max(s_1^2, s_0^2)}{\min(s_1^2, s_0^2)} \geq 1, \tag{3}$$

always at least 1 regardless of which group’s variance is larger, with $VR^* > 2$ indicating imbalance under the same convention. On the log scale, $\log(VR^*) = |\log(s_1^2) - \log(s_0^2)|$: the symmetrized ratio is exactly the exponentiated absolute difference in log-variances between the two groups, an identity that motivates the k -sample construction below.

The log scale is not merely an algebraic convenience. Rubin’s VR is a multiplicative, not additive, quantity: a ratio of 2 and a ratio of 0.5 represent the identical magnitude of imbalance in opposite directions, which is precisely why VR^* symmetrizes by taking a max/min ratio rather than a difference. Taking logs converts this multiplicative, reciprocal-symmetric quantity into an additive, sign-symmetric one ($\log 2 = -\log 0.5$), so that ordinary Euclidean dispersion — the same root-mean-square construction Cohen’s f uses for means — becomes the natural k -sample generalization. A k -sample statistic built directly from the raw variances or their ratios would not have this property: ordinary dispersion of variances themselves, or of ratios of variances, does not treat a doubling and a halving as equal-magnitude departures from balance, exactly the asymmetry VR^* ’s own construction was designed to avoid at $k = 2$.

2.3 The k -Sample Extension: F_{VR}

Let $\ell_j = \log(s_j^2)$ denote Group j 's log-variance, and $w_j = n_j/N$, the same group-size weights used for Cohen's f (Linden, 2026a). These are deliberately raw group-membership proportions, not a post-weighting measure such as effective sample size under w_{ji} : the omnibus statistic aggregates *across* groups by each group's share of the original sample, while each group's own variance s_j^2 is separately computed using whatever observation-level weights w_{ji} the analysis specifies (Section 2.1). After weighting, each group's variance s_j^2 is computed in the reweighted pseudo-population; its contribution to the omnibus F_{VR} , however, remains proportional to its original sample size rather than its weighted mass. We adopt this for consistency with Cohen's f , where the identical choice is made for the identical reason. An investigator whose estimand instead assigns each group's contribution in proportion to its weighted mass, or wants every pairwise contrast to carry equal importance regardless of group size, should consult the per-group and pairwise decomposition directly (Section 2.4) rather than rely on this aggregation choice.

This is a substantive choice about the estimand, not an incidental detail. These are size weights defining the omnibus estimand, not inverse-variance or precision weights intended to optimize statistical precision; n_j/N is not formally a precision weight for ℓ_j , particularly under non-normality or observation-level weighting, and should not be read as an attempt to account for the relative precision of each group's variance estimate. Section 3.4 reports a separate, empirical observation that this size-weighting may also confer some robustness against noisy small groups in one simulation design; that is a distinct, empirical finding, not a property built into the definition here. We define

$$F_{VR} = \sqrt{\sum_{j=1}^k w_j (\ell_j - \bar{\ell})^2}, \quad \bar{\ell} = \sum_{j=1}^k w_j \ell_j, \quad (4)$$

a size-weighted root-mean-square of each group's log-variance deviation from the weighted mean log-variance, exactly analogous in construction to Cohen's f for means (Linden, 2026a),

with log-variance playing the role that the group mean itself plays there.

Backward compatibility at $k = 2$. By the same algebra used to establish $f = \sqrt{w_1 w_2} |d|$ for Cohen’s f (Linden, 2026a) — an argument that depends only on the fact that ℓ_1, ℓ_2 are two real numbers with a weighted mean $\bar{\ell} = w_1 \ell_1 + w_2 \ell_2$, not on anything specific to means as opposed to log-variances — F_{VR} reduces exactly to

$$F_{VR} = \sqrt{w_1 w_2} |\log(VR^*)| \quad (5)$$

at $k = 2$, for any finite sample, not merely asymptotically. This identity was confirmed both algebraically and against real `varatio` output on unequal group sizes ($n_1 = 2000$, $n_2 = 1000$; $VR^* = 5.155$), matching to four decimal places.

2.4 Per-Group and Pairwise Decomposition

Each group’s deviation $\ell_j - \bar{\ell}$ reports how far that group’s own log-variance sits from the weighted mean log-variance, the same log-variance units contributing to the omnibus $F_{VR}^2 = \sum_j w_j (\ell_j - \bar{\ell})^2$, so that an investigator can identify which specific group is driving a large F_{VR} . Because SD_p -style pooling is not required here — each ℓ_j is estimated from its own group’s data, with no shared standardizer across groups — pairwise comparisons reduce directly to the two-group construction of Section 2.2: for any two groups j, j' , $VR_{jj'}^* = \exp(|\ell_j - \ell_{j'}|)$, giving $\binom{k}{2}$ pairwise ratios computed from the same per-group log-variances underlying the omnibus statistic. As with the corresponding per-level statistic for means (Linden, 2026a), we report point estimates for both the per-group and pairwise quantities; no closed-form confidence interval is provided, for reasons discussed in Section 5.4, where a bootstrap alternative is recommended instead.

2.5 Benchmark: GMVR and Maximum Pairwise Variance Ratio

We compare F_{VR} against two natural comparators based on existing aggregation practices, computed from the identical per-group log-variances: the geometric mean variance ratio,

$$GMVR = \exp\left(\binom{k}{2}^{-1} \sum_{j < j'} |\log(VR_{jj'}^*)|\right), \quad (6)$$

adapting Linden and Samuels (2013)’s use of the geometric mean of pairwise variance ratios across *covariates* to an average across *group pairs* instead — a repurposing of an existing idea to a new setting, not itself a previously established k -group statistic; and the maximum pairwise ratio, $\max_{j < j'} VR_{jj'}^*$, which is a genuinely established software practice, the aggregation logic already used by the R package `cobalt` for multi-category treatments, adapted from the maximum-standardized-bias practice of Lopez and Gutman (2017).

Relationship between F_{VR} , GMVR, and the maximum. Because each $\ell_j - \bar{\ell}$ is a deviation from a weighted mean, $\sum_j w_j(\ell_j - \bar{\ell}) = 0$, and by the same identity used for Cohen’s f (Linden, 2026a), $F_{VR}^2 = \sum_{j < j'} w_j w_{j'} [\log(VR_{jj'}^*)]^2$. Thus, F_{VR} is a size-weighted quadratic combination of the same pairwise log-ratios that GMVR averages linearly and without group-size weights. The maximum statistic retains only the single worst pair. In norm terminology, GMVR is a monotone transformation of an unweighted L_1 -type aggregate (specifically, $\exp$ of the mean absolute log-ratio, not itself proportional to the L_1 norm on the original ratio scale), the maximum statistic corresponds to L_∞ , and F_{VR} is a size-weighted L_2 -type aggregate. The latter is not literally the conventional L_2 norm because the $w_j w_{j'}$ weighting differs from an unweighted norm of the pairwise vector. This parallels the norm-based framing developed for the corresponding mean-balance statistics (Linden, 2026a).

Throughout this section, $\overline{|\log VR^*|}$ denotes the *arithmetic mean* of the $\binom{k}{2}$ pairwise absolute log-ratios — the same quantity underlying GMVR (Section 2.5). It does not denote the range or the single largest pairwise value. These quantities have different minimizing configurations, and Proposition 1 below concerns the mean specifically, denoted with the explicit

overline throughout to avoid ambiguity with the maximum-based statistic of Section 2.5.

Proposition 1. *Under equal group weights ($w_j = 1/k$ for all j), the ratio $F_{VR}/\overline{|\log VR^*|}$, where $\overline{|\log VR^*|}$ is the arithmetic mean of the $\binom{k}{2}$ pairwise absolute log-variance-ratios, attains its minimum over all attainable configurations of k log-variances at equally spaced log-variances, where*

$$\frac{F_{VR}}{\overline{|\log VR^*|}} = \sqrt{\frac{3(k-1)}{4(k+1)}}. \quad (7)$$

This is the identical result proven for Cohen’s f (Linden, 2026a, Proposition 1), and the proof requires no modification: it depends only on $\ell_1, \dots, \ell_k$ being k real numbers with a well-defined weighted mean and a Cauchy–Schwarz argument over their deviations, using nothing specific to means as opposed to log-variances. We confirmed this transfer both by direct substitution into the existing proof and by numerical search, at two independent scales: for $k = 3, 4, 6$, minimizing $F_{VR}/\overline{|\log VR^*|}$ over 10^5 , and separately over 5×10^5 , randomly generated log-variance configurations at each k converged to 0.6124, 0.6708, and 0.7320 respectively in both searches, matching the closed form to four decimal places. This minimum should not be confused with the (smaller) minimum of F_{VR} divided by the *range* of the log-variances (equivalently, the single largest pairwise ratio), which is a different quantity, governed by a different extremal configuration, and is not the subject of this proposition.

2.6 Simulation Study Design

Before describing the simulation design, it is useful to distinguish our setting from Rubin’s. All three statistics are designed to detect variance imbalance that may matter for estimation bias. Rubin (Rubin, 2001; Cochran and Rubin, 1973) studied whether variance imbalance signals failure of subsequent regression adjustment; our simulation instead makes variance imbalance the direct channel producing bias. The simulation therefore evaluates the latter mechanism, not Rubin’s original regression-adjustment setting.

We evaluated F_{VR} , GMVR, and the maximum pairwise ratio against actual downstream

estimation bias across a Monte Carlo study spanning five distinct components, summarized in Table 1: a primary mechanism, three structurally distinct variants testing generalizability beyond that one mechanism, and a sample-size sensitivity check.

2.6.1 Isolating a Variance-Driven Bias Channel

Unlike mean imbalance, whose relationship to downstream bias is well-established, variance imbalance’s relationship to bias required a purpose-built data-generating process: a covariate whose *mean* is balanced across groups by construction, so that any resulting bias can only flow through variance. In the primary mechanism, $X \mid S \sim N(0, \sigma(S)^2)$ for a latent driver $S \sim \text{Uniform}(0, 1)$ with $\sigma(S) = 1 + 3S$, giving X mean zero in every group regardless of how treatment assignment depends on S . The true outcome model, $Y = \mu_T + \beta_3 X^2 + \varepsilon$, includes a quadratic term in X ; the analyst’s naive model regresses Y on treatment indicators and X linearly, omitting X^2 . Because $E[X \mid T = j] = 0$ in every group, this omitted-variable bias is driven entirely by between-group differences in $E[X^2 \mid T = j] = \text{Var}(X \mid T = j)$, not by X ’s mean — isolating a variance-driven bias channel that a mean-based diagnostic could not detect, and that F_{VR} , GMVR, and the maximum ratio are specifically built to.

Treatment assignment follows a random-utility multinomial model in S , confounding strength scaled by $\gamma \in \{0, 0.25, 0.5, 1, 1.5\}$, crossed with the same equal/unequal group-size balance mechanism used in the mean-based study (Linden, 2026a). Generalized propensity score models were fit under two specifications: correct, including S ; and misspecified, using a noisy proxy for S . Bias was computed as the naive model’s estimated treatment contrast (each group relative to a reference group, canceling a constant term common to every group’s raw predicted level) minus the true parameter contrast, using the replicate’s own sample data throughout.

2.6.2 Calibration

Because the omitted-quadratic-term bias mechanism can saturate under a large enough effect size, giving no discriminating power for threshold-finding once bias exceeds conventional thresholds regardless of confounding strength, β_3 was calibrated so that the no-confounding baseline sits near a 0.10-SD bias threshold and maximum confounding approaches a 0.20-SD threshold, using the outcome noise term’s known unit standard deviation as the reference scale. The primary mechanism, generalizability Variant B, and the sample-size sensitivity check share $\beta_3 = 0.05$; Variants A and C required separate calibration ($\beta_3 = 0.30$ and 0.40 respectively) given their distinct bias-generating mechanisms, described next.

2.6.3 Generalizability Variants

Three structurally distinct mechanisms tested whether F_{VR} ’s relationship to bias depends on the specific way variance imbalance was constructed. **Variant A** replaces the omitted quadratic term with an omitted mean absolute deviation, $Y = \mu_T + \beta_3|X| + \varepsilon$: for normal X , $E|X| = \sigma\sqrt{2/\pi}$, linear in σ rather than quadratic, testing a genuinely different functional dependence on variance. **Variant B** introduces a second, independently variance-imbalanced covariate X_2 , driven by an independent latent driver S_2 . Both X_1^2 and X_2^2 are omitted from the naive model, while F_{VR} is computed on X_1 alone. This tests whether F_{VR} remains predictive of total bias when another source of variance imbalance is not represented in the diagnostic — a realistic robustness check, since applied settings rarely have exactly one confounder. **Variant C** draws X from a mean-centered lognormal distribution rather than Normal, testing sensitivity to non-normal covariate shape; mean-centering used the exact lognormal mean formula, $X = \exp(\sigma(S)Z) - \exp(\sigma(S)^2/2)$ for $Z \sim N(0, 1)$, so that $E[X | T = j] = 0$ is preserved exactly despite the skewed shape.

2.6.4 Sample-Size Sensitivity

The primary mechanism was additionally re-run at $k = 3, 4, 6$ across $N \in \{200, 2000, 10,000\}$ (in addition to the primary design’s own $N \in \{500, 2000\}$), to test whether the relationship between F_{VR} and bias, and any candidate interpretive threshold, is stable across sample size rather than specific to one N .

2.6.5 Common Design Elements

Each (k, N) combination in every component was replicated 1000 times per cell across five γ levels and two balance scenarios (10 cells), under three weighting arms (unweighted, correctly weighted, misspecified), for 10,000 replicates per file. The resulting design comprised approximately 720,000 replicates across all five components; Table 1 summarizes their allocation by mechanism, k , and N . All analyses were conducted in Stata (Version 19) using the community-contributed command `varatio` (Linden, 2026b).

3 Results

3.1 Correlation with Downstream Bias

In the primary mechanism, F_{VR} and GMVR tracked mean absolute bias comparably at $k = 3$ ($r = 0.529$ and 0.537), with the maximum statistic close behind ($r = 0.530$). At $k = 6$, the correlations were 0.423 , 0.471 , and 0.349 , respectively.

The weaker performance of the maximum statistic is consistent with greater sensitivity to sampling variability when only the worst pair is retained, matching the same weakness documented for mean imbalance (Linden, 2026a). The simulation does not directly establish that mechanism: the $k = 6$, $N = 200$ result of Section 3.4, where the maximum statistic’s correlation with bias collapses most severely, is consistent with this interpretation without demonstrating the underlying variance mechanism directly. This ranking held, with the same qualitative pattern, across all three generalizability variants (Section 3.3).

3.2 Threshold-Finding: Logistic Crossing Points

Using mean absolute bias exceeding 0.10 SD as ground truth for “unacceptable” imbalance, we located, for each mechanism and k , the F_{VR} value at which a fitted logistic model predicts a 50% chance of exceeding that threshold. Pooled across k , this crossing point was 0.141 for the primary mechanism, 0.157 for Variant A, 0.108 for Variant B, and 0.291 for Variant C (pooled across all four: 0.175) — a real, mechanism-dependent range rather than a single stable number, but one concentrated between roughly 0.11 and 0.29 across four structurally distinct bias-generating mechanisms.

3.3 Generalizability Across Mechanisms

Variant C’s elevated crossing point is consistent with its weaker overall correlation with bias ($r = 0.634$ pooled, versus $r > 0.89$ for the other two variants): higher kurtosis in the skewed covariate increases sampling variability in both F_{VR} itself and in the resulting bias estimate simultaneously, attenuating the observed relationship without indicating that the underlying mechanism differs. Variant B confirmed that F_{VR} , computed on one covariate alone, remained strongly correlated with total bias arising from two independent, unobserved sources of imbalance ($r = 0.900$) in this particular data-generating process. This does not show that F_{VR} detects unseen confounding in general; it shows that, in this DGP, the statistic computed on one observed covariate correlated with total downstream bias even when a second, unobserved variance-imbalanced covariate also contributed to that bias — a more limited, but still practically relevant, claim, since applied settings rarely have exactly one confounder.

3.4 Sample-Size Sensitivity

The absolute-bias-threshold framework itself proved sensitive to sample size in a way worth reporting plainly rather than minimizing. At $N = 200$, logistic crossing points for the

primary mechanism, re-run across $k = 3, 4, 6$, were negative or near zero (-0.022 at $k = 3$; -0.027 at $k = 4$), meaning the fitted model predicted better-than-even odds of exceeding the bias threshold even at $F_{VR} \approx 0$; crossing points stabilized to the primary design’s own range only at $N = 2000$ and above. This reflects the noise floor inherent in bias itself, driven by ordinary regression-estimation variance that scales with N , separately from the systematic, variance-driven signal F_{VR} is built to detect; a fixed absolute bias threshold (0.10 SD) therefore does not correspond to a stable point on the F_{VR} scale once N varies widely, a limitation discussed further in Section 5.4.

A notable finding for F_{VR} specifically emerged when data spanning $N = 200$ to 10,000 were pooled at $k = 6$: GMVR’s and the maximum ratio’s correlation with bias collapsed to near zero ($r = 0.020$ and 0.019), while F_{VR} retained substantial predictive power ($r = 0.638$). Because this pooled figure could in principle reflect heterogeneity across the three widely different N values rather than a genuine within- N advantage, we report the correlation separately at each (k, N) combination in Table 3. The dramatic divergence is concentrated in the $k = 6$, $N = 200$ cell, where GMVR’s and the maximum ratio’s correlation with bias collapse to near zero ($r = 0.022$ and 0.025) while F_{VR} retains real signal ($r = 0.378$). At $N = 200$, F_{VR} also has a smaller advantage at $k = 3$ and $k = 4$. Relative to the maximum statistic, the correlation differences are $+0.084$ and $+0.163$, respectively; relative to GMVR, both are approximately $+0.02$. At $N \geq 2000$, differences among the three statistics become small regardless of k . The visible pattern — F_{VR} ’s advantage over the maximum statistic growing as observations per group become scarcer — is consistent across all three k values examined, though nine cells are not enough to claim a formal monotonic relationship. This confirms the advantage is a genuine within- N phenomenon rather than a pooling artifact — the question the pooled figure alone could not answer — while showing it is not confined to the single most extreme cell.

3.5 Classification Accuracy

Treating mean absolute bias ≤ 0.10 as ground truth for “truly balanced,” we assessed how often each statistic’s own classification (balanced/imbalanced, using Rubin’s fixed $VR > 2$ convention for GMVR and the maximum ratio, and four candidate cutoffs for F_{VR}) matched ground truth, pooled across all mechanisms and k . GMVR, using Rubin’s convention, achieved 98.9% specificity (correctly clearing truly balanced cases) but only 3–17% sensitivity across mechanisms (correctly flagging truly imbalanced cases): Rubin’s threshold, calibrated to detect the severe regression-adjustment breakdown documented by Cochran and Rubin (Cochran and Rubin, 1973), essentially never triggers against a substantially milder 0.10-SD bias standard. F_{VR} at 0.15 achieved the highest overall accuracy of any statistic or cutoff tested (68.5%), compared with 66.0% for the maximum ratio and 51.9% for GMVR. Among the four F_{VR} cutoffs, accuracies were 67.0%, 68.5%, 65.6%, and 56.2% at 0.10, 0.15, 0.20, and 0.30, respectively. This supports 0.15 as a useful candidate classification cutoff, but not as independently validated: it was one of the candidates evaluated on these same simulation data. Although 0.15 performed best as a classification cutoff among the four candidates examined, it should not be interpreted as a universal boundary for acceptable balance: Section 5.3 shows this same value already exceeds the 50% crossing point for two of the four mechanism-specific thresholds and sits essentially at a third, so a classification-optimal cutoff and an “acceptable” threshold are related but distinct questions, and the two should not be conflated.

Classification performance above was pooled across mechanisms and therefore does not characterize mechanism-specific sensitivity and specificity.

4 Applied Example

4.1 Study Context and Data

We illustrate F_{VR} , GMVR, and the maximum pairwise ratio using data from a disease management (DM) program for patients with congestive heart failure examined in two prior methodological papers introducing and comparing multivalued-treatment adjustment approaches on this dataset (Linden, 2014; Linden et al., 2016). The program was implemented in a large health plan in the western United States; enrolled patients received one of two interventions based on a program nurse’s subjective assessment of patient needs and preferences: periodic telephone calls from a nurse to discuss self-management behaviors, or remote tele-monitoring (RTM) involving daily electronic transmission of disease-related symptoms with nurse follow-up when indicated (Linden et al., 2006). Health plan members with the condition who did not enroll served as a non-participant comparison group. The data comprise 6612 non-participants (Control), 654 telephonic participants (Calls), and 705 RTM participants ($N = 7971$), each with pre-intervention data.

4.2 Covariate Balance Assessment

Following Linden (2014) and Linden et al. (2016), MMWS weights were constructed using the generalized approach for nominal treatments described in Linden (2014): a generalized propensity score for each of the three treatment levels was estimated by multinomial logistic regression. We apply F_{VR} , GMVR, and the maximum pairwise ratio to a medical risk score reflecting each patient’s model-predicted near-term medical costs based on prior health care utilization history, before and after MMWS weighting.

4.3 Results

Table 6 reports all three statistics, together with the per-group and pairwise decomposition, before and after weighting. Before weighting, the omnibus $F_{VR} = 0.182$ sits in the region this

paper describes as requiring the decomposition rather than the omnibus number alone (Section 5.3). GMVR (1.584) and the maximum pairwise ratio (1.994) tell a consistent story on their own scales; the Calls-versus-RTM pair sits almost exactly at Rubin’s $VR^* > 2$ convention. The decomposition identifies the source: RTM’s variance (138.74) differs sharply from Control’s (260.85) and Calls’s (276.62), and despite RTM holding only 8.8% of the sample’s weight, its large deviation dominates F_{VR} — a real-data illustration of the masking mechanism confirmed by direct simulation in Section 5.4, where a minority group’s imbalance is discounted only when its magnitude is comparable to the other groups’, not unconditionally. After MMWS weighting, all three statistics improved substantially ($F_{VR} = 0.091$, within the region characterized here as reasonably reassuring; GMVR= 1.224; maximum ratio= 1.354), with every pairwise comparison falling within a narrow band ($VR^* = 1.15$ to 1.35), consistent with a genuine reduction in imbalance across every pair rather than a reduction in one dominant discrepancy. All three statistics agree on the substantive conclusion despite differing scales and despite baseline variance imbalance being concentrated specifically in the RTM arm: all three statistics therefore indicate a substantial reduction in variance imbalance after weighting.

5 Discussion

5.1 Principal Findings

Four findings emerge.

First, F_{VR} and GMVR generally tracked downstream estimation bias at least as well as the maximum pairwise ratio, with GMVR often showing the highest correlation, replicating for variance a broadly similar comparative ranking to the one already established for mean imbalance (Linden, 2026a).

Second, this ranking held across four structurally distinct bias-generating mechanisms — an omitted quadratic term, an omitted mean absolute deviation, a second unobserved

imbalanced covariate, and a skewed covariate distribution — though the specific numeric threshold varied meaningfully across them (0.108 to 0.291), evidence of real generalization rather than an artifact of one particular design.

Third, N -specific correlations (Table 3) confirm that GMVR’s and the maximum ratio’s near-total loss of signal at $k = 6$ (Section 3.4) is a genuine within- N phenomenon, not an artifact of pooling across sample sizes: the most dramatic divergence occurs at $k = 6$, $N = 200$ (six groups sharing only 200 observations), where GMVR and the maximum ratio collapse to near-zero correlation while F_{VR} retains real signal, and a smaller but real advantage over the maximum statistic specifically is visible at $N = 200$ for $k = 3, 4$ as well, with all three statistics converging as N grows.

Fourth, a classification-accuracy analysis against actual bias identified $F_{VR} \leq 0.15$ as achieving the best overall accuracy of any statistic or cutoff tested among four prespecified candidates, additional supporting evidence for a candidate classification cutoff, though not independent validation, since 0.15 was itself one of the cutoffs evaluated on the same simulation data.

5.2 Why the Threshold Isn’t Sharper

The candidate range (0.108–0.291 across mechanisms) is wider than a single number, and this reflects a real property of the problem, not a weakness of the analysis. Rubin’s own two-group convention was established once, for one adjustment method (linear regression) and one severity criterion (gross bias correction failure); this paper’s range was derived across four different ways variance imbalance can translate into bias, and the specific value at which a fixed 0.10-SD bias threshold is crossed depends, correctly, on how strongly and how noisily that particular mechanism connects variance to bias. A single, mechanism-free number would misrepresent this dependence rather than resolve it.

5.3 Practical Advice

Based on this evidence, we suggest three interpretive regions as a starting point, not as validated thresholds in Rubin’s sense. F_{VR} below approximately 0.10 was reasonably reassuring across every mechanism and sample size examined here. F_{VR} above approximately 0.30 indicated a real concern in nearly every setting tested. Values from approximately 0.10 to 0.30 were genuinely indeterminate and context-dependent; an investigator with the ability to simulate a design resembling their own application should derive a threshold directly from that design’s own bias-correlation relationship, following the approach of Section 2.6, rather than import this paper’s range uncritically.

These three regions do not collapse to a single “acceptable” cutoff at 0.15. The four mechanism-specific crossing points (0.108, 0.141, 0.157, 0.291; Section 3.2) already show why: 0.15 exceeds the 50% crossing point for two of the four mechanisms and sits essentially at the crossing point for a third, comfortably below it only for Variant C. The 0.10–0.30 region spans exactly this range of mechanism-specific crossing points, which is why it should not be treated as either acceptable or concerning by default.

For a rough sense of what a given F_{VR} value corresponds to on the more familiar two-group ratio scale, the exact $k = 2$ identity of Section 2.3 can be read in reverse: for equal group sizes, $F_{VR} = \frac{1}{2}|\log VR^*|$, so $F_{VR} = 0.15$ corresponds to $VR^* = e^{0.30} \approx 1.35$, roughly a 35% variance difference between the two groups in the equal-sized two-group case specifically; this identity does not extend to $k > 2$, where $F_{VR} = 0.15$ does not imply that the largest pairwise ratio is 1.35. More generally, for unequal weights in the $k = 2$ case, $VR^* = \exp(F_{VR}/\sqrt{w_1 w_2})$. This is offered as an interpretive bridge, not a new result, since it follows directly from the identity already established in Section 2.3.

As with Cohen’s f (Linden, 2026a), F_{VR} should be reported alongside its per-group and pairwise decomposition, not as a single omnibus number in isolation, and its n_j/N weighting means that imbalance concentrated in a small treatment group will be underrepresented in the omnibus statistic relative to imbalance of the same magnitude in a large group — the

identical substantive consideration already raised for Cohen’s f (Linden, 2026a), and confirmed directly for variance imbalance in a masking-mechanism check reported in Section 5.4.

5.4 Limitations

Several limitations qualify these findings. First, F_{VR} carries no closed-form confidence interval in this implementation: a naive generalization of the unweighted variance-of-log-variance formula (itself verified, empirically, to $\text{Var}(\log s^2) \approx (\kappa - 1)/n$ for kurtosis κ , across distributions from uniform to exponential) is not valid once weights correlate with the covariate, exactly the condition inverse-probability weighting is designed to produce. Rather than report an invalid formula, `varatio` returns every quantity it computes — omnibus, per-group, and pairwise — so that any of them can be bootstrapped directly; users requiring a formal interval should budget for that additional computation.

Second, the masking mechanism — imbalance concentrated in a small group contributing less to F_{VR} than the same imbalance in a large group — was confirmed directly for variance: moving an identical variance discrepancy from a minority group ($n=50$ of 3050) to a majority group ($n=1000$) increased F_{VR} nearly fourfold (0.254 to 0.958) while GMVR and the maximum ratio moved by only 6–8%, since neither incorporates group-size weighting. This is the intended behavior of a size-weighted statistic, not a defect, but it means F_{VR} should not be relied upon in isolation whenever a contrast involving a specific, less-common treatment arm is of direct substantive interest.

Third, the maximum pairwise ratio showed real fragility at higher k , particularly under correctly specified weighting: its correlation with bias fell to $r = 0.097$ in one Variant B condition ($k = 6$, correct GPS), and one logistic crossing point for Variant C at $k = 6$ was itself negative (-0.008), both plausibly driven by extreme-outlier instability in near-empty groups under strong confounding and unequal balance. This is a limitation of the maximum-based statistic specifically, not of F_{VR} , but it means comparisons involving the maximum ratio at higher k should be interpreted cautiously.

Fourth, the absolute-bias threshold framework was not stable across widely varying sample sizes: crossing points were negative or near zero at $N = 200$. The candidate range in Section 5.3 should therefore be applied cautiously when treatment groups contain only hundreds of observations, since it was derived predominantly from designs with $N \geq 500$.

Fifth, each generalizability variant tested a single fixed effect-size parameter, calibrated specifically to produce discriminating power for threshold-finding rather than chosen to represent any particular applied setting; whether the comparative ranking and candidate range are stable across a wider range of effect sizes within each mechanism was not assessed.

5.5 Conclusion

Variance imbalance across more than two treatment groups has lacked a purpose-built omnibus diagnostic. F_{VR} fills that gap with an exact relationship to Rubin’s VR at $k = 2$ and a mathematical structure parallel to Cohen’s f ’s relationship to the standardized mean difference. Across four bias-generating mechanisms, F_{VR} tracked downstream estimation bias comparably to existing aggregation approaches and retained more signal when many groups shared very few observations, confirmed by correlations reported separately within each sample-size stratum rather than pooled across them. The candidate interpretive range offered here is a starting point, grounded in the same simulation-based logic that established Rubin’s convention five decades ago, not a claim to the same accumulated authority. As with Cohen’s f , F_{VR} is best reported alongside its full per-group and pairwise decomposition. Investigators able to simulate a design resembling their own application should derive an application-specific threshold directly.

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Table 1: Simulation Study Design

Component	Description
Primary mechanism	Omitted quadratic term ($Y = \mu_T + \beta_3 X^2 + \varepsilon$); $\beta_3 = 0.05$; $k \in \{3, 4, 6\}$, $N \in \{500, 2000\}$; 60,000 replicates
Variant A	Omitted mean absolute deviation ($Y = \mu_T + \beta_3 X + \varepsilon$); $\beta_3 = 0.30$; $k \in \{3, 4, 6\}$, $N = 2000$; 90,000 replicates
Variant B	Two independent variance-imbalanced covariates, F_{VR} computed on one alone; $\beta_3 = 0.05$; $k \in \{3, 4, 6\}$, $N = 2000$; 90,000 replicates
Variant C	Mean-centered lognormal covariate (skewed); $\beta_3 = 0.40$; $k \in \{3, 4, 6\}$, $N = 2000$; 90,000 replicates
Sample-size check	Primary mechanism; $k \in \{3, 4, 6\}$, $N \in \{200, 2000, 10,000\}$; 90,000 replicates
Confounding strength (γ)	0 (none), 0.25, 0.5, 1, 1.5 (strong)
Group-size balance	Equal: all k groups share the same baseline assignment probability. Unequal: group-specific intercepts produce a skewed baseline split.
GPS weighting	Unweighted; correctly specified (includes latent driver S); misspecified (noisy proxy for S)

Note: each (k, N) combination replicated 1000 times per cell across the 10 $\gamma \times$ balance cells, under all three weighting arms, for approximately 720,000 replicates in total across all five components. All analyses conducted in Stata using the community-contributed command `varatio`.

Table 2: Correlation with actual estimation bias, by mechanism and k

Mechanism	k	F_{VR}	GMVR	Max VR^*
Primary	3	0.529	0.537	0.530
	4	0.487	0.507	0.359
	6	0.423	0.471	0.349
Variant A	3	0.686	0.703	0.700
	4	0.606	0.633	0.629
	6	0.456	0.513	0.512
Variant B	3	0.591	0.645	0.636
	4	0.503	0.574	0.486
	6	0.450	0.535	0.107
Variant C	3	0.470	0.499	0.489
	4	0.403	0.444	0.427
	6	0.321	0.405	0.410

Note: correlation with mean absolute bias, pooled across γ , balance, weighting arm, and (for the primary mechanism) N .

Table 3: Correlation with actual estimation bias, by k and N (nsens study, primary mechanism)

k	N	F_{VR}	GMVR	Max VR^*
3	200	0.810	0.788	0.726
	2000	0.966	0.960	0.950
	10000	0.992	0.981	0.974
4	200	0.703	0.684	0.540
	2000	0.912	0.915	0.904
	10000	0.976	0.961	0.949
6	200	0.378	0.022	0.025
	2000	0.736	0.757	0.739
	10000	0.834	0.826	0.826

Note: correlation with mean absolute bias, pooled across γ , balance, and weighting arm within each (k, N) cell; not pooled across N , unlike Table 2. The $k = 6$, $N = 200$ row (bolded) shows the most dramatic divergence; F_{VR} also retains a smaller but real advantage over the maximum statistic at $N = 200$ for $k = 3, 4$, with all three statistics converging at $N \geq 2000$.

Table 4: Logistic-regression crossing points: F_{VR} value at which predicted $P(\text{bias} > 0.10) = 0.5$

Mechanism	$k = 3$	$k = 4$	$k = 6$
Primary	0.141	0.136	0.124
Variant A	0.157	0.160	0.161
Variant B	0.108	0.113	0.115
Variant C	0.291	0.290	0.279

Note: pooled across weighting arm and balance scenario within each mechanism/ k cell.

Table 5: Classification accuracy against actual bias (mean bias ≤ 0.10 as ground truth for “balanced”), pooled across all mechanisms and k

Statistic (cutoff)	Sensitivity	Specificity	Accuracy	False-negative rate
GMVR (2.0)	0.087	0.989	0.519	0.913
Max VR^* (2.0)	0.464	0.889	0.660	0.536
F_{VR} (0.10)	0.826	0.553	0.670	0.174
F_{VR} (0.15)	0.675	0.741	0.685	0.325
F_{VR} (0.20)	0.511	0.845	0.656	0.489
F_{VR} (0.30)	0.215	0.951	0.562	0.785

Note: sensitivity = correctly flags real imbalance; specificity = correctly clears real balance.
Pooled across primary, Variant A, Variant B, Variant C, and the sample-size check.

Table 6: Applied example: predictive model score variance-ratio balance before and after MMWS weighting

	Unweighted	MMWS-weighted
F_{VR} (omnibus)	0.182	0.091
GMVR	1.584	1.224
Max VR^*	1.994	1.354
Per-group variance		
Control (N=6612)	260.85	259.77
Calls (N=654)	276.62	226.83
RTM (N=705)	138.74	191.82
Pairwise VR^*		
Control vs. Calls	1.061	1.145
Control vs. RTM	1.880	1.354
Calls vs. RTM	1.994	1.183

Note: $N = 7971$ (6612 Control, 654 Calls, 705 RTM). GMVR and max VR^* computed from the same three pairwise VR^* 's reported below.